

- 5 Phosphate ions are absorbed from the soil solution by roots and are needed for cellular processes throughout plants.

Scientists investigated the movement of phosphate ions in flowering plants. The scientists discovered that phosphate ions in the leaves are transported from the roots in the xylem. Only a small proportion of the phosphate ions that are absorbed are transported to the growing points of the roots and shoots.

- (a) Suggest why only a small proportion of the absorbed phosphate ions are transported to the growing points.

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- (b) *Gossypium hirsutum* is the most common species of plant grown for the production of cotton across the world.

Scientists carried out an investigation to trace the pathway taken by phosphate ions from the leaves of cotton plants into the stems. The scientists used a radioactive isotope of phosphorus ( $^{32}\text{P}$ ) to trace the pathway of phosphate ions.

Some cotton plants were divided into two groups: **A** and **B**.

In group **A**, the scientists:

- inserted impermeable waxed paper between the xylem and phloem in the stem below a leaf of each plant
- injected a solution containing phosphate ions labelled with  $^{32}\text{P}$  (labelled phosphate ions) into a vein of each leaf, as shown in Fig. 5.1.

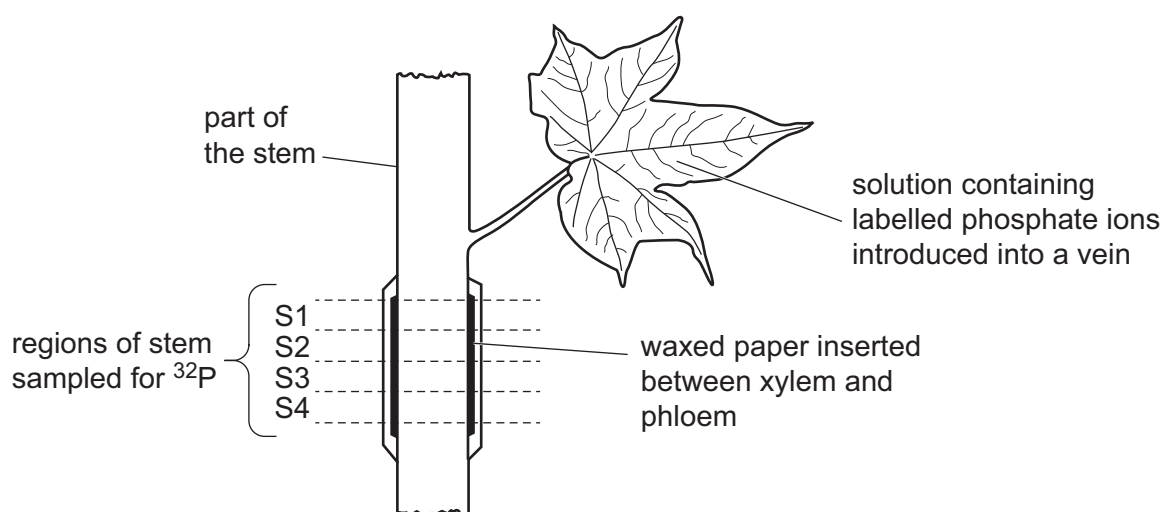


Fig. 5.1

The procedure was repeated on the plants in group **B** but without inserting the waxed paper.

After one hour, the scientists determined the percentage of labelled phosphate ions in the four sections of the stem, **S1** to **S4**, shown in Fig. 5.1. The results are shown in Table 5.1.

**Table 5.1**

| region of stem sampled | percentage of injected labelled phosphate ions in stem tissues |       |                                     |       |
|------------------------|----------------------------------------------------------------|-------|-------------------------------------|-------|
|                        | group A – stems with waxed paper                               |       | group B – stems with no waxed paper |       |
|                        | phloem                                                         | xylem | phloem                              | xylem |
| <b>S1</b>              | 12                                                             | 1     | 15                                  | 5     |
| <b>S2</b>              | 7                                                              | <1    | 10                                  | 6     |
| <b>S3</b>              | 13                                                             | 0     | 5                                   | 2     |
| <b>S4</b>              | 5                                                              | <1    | 3                                   | 1     |

Use Fig. 5.1 and the data in Table 5.1 to discuss the pathway taken by the solution containing phosphate ions labelled with  $^{32}\text{P}$  in cotton plants.

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[Total: 6]